



A Place to Learn, A Place to Grow

CAT | CLAT | IPMAT | MBA CET | MCA CET | CRT

IPMAT INDORE 2020

QUESTION PAPER WITH SOLUTIONS



Quantitative Ability SA

1. In a division problem, product of quotient and there mainderis 24while their sum is 10.If the divisoris 5 then dividend is

2. The shortest distance from the point $(-4,3)$ to the circle $x^2 + y^2 = 1$ is

3. The value of $0.04^{\log \sqrt{5}} \left(\frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \right)$ is _____

4. Suppose $\begin{bmatrix} a & a^2 & b^2 & -1 \\ b & b^2 & b^3 & -1 \\ c & c^2 & c^3 & -1 \end{bmatrix} = 0$. where a, b and c care distince real numbers. If a = 3, then the value of abc is _____

5. The $f(x)$ minimum value of $=|3-|+|2+|2|+|5-|$ lis equal to _____

6. Ashok purchased pens and pencils in the ratio 2:3 during his first visit and paid Rs. 86 to the shopkeeper. During his second visit,he purchased pens and pencils in the ratio 4:1 and paid Rs.112. The cost of a pen as well as a pencil in rupees is apositive integer.If Ashokpurchased four pens during his second visit,then the amount he paid in rupees for the pens during the second visit is_____

7. In a four-digit number, the product of thousands digit and units digit is zero while their difference is 7 Product of the middle digits is 18.The thousands digit is as much more than theunits digit as the hundreds digit is more than the tens digit. The four-digit number is_____

8. Out of 80 students who appeared for the school exams in Mathematics (M),Physics (P) and Chemistry (C) 50 passed M,30passed Pand40passed C.At most 20 students passed M and P,at most 20 students passed P and C and at most 20 students passed C and M.Themaximum number of students who could have passed all three exams is _____

9. Two friends run a 3 kilometer race along a circular course of length 300 meters.
If their speeds are in ratio3:2,the number of times the winner passes the other is_____

10. Out of 13 objects, 4 are indistinguishable and rest are distinct. The number of ways we can choose 4 objects out of 13 objects is _____

Quantitative Ability MCQ

11. The probability that a randomly chosen factor of 10^{19} is a multiple of 10^{15} is
- (a) $\frac{1}{25}$ (b) $\frac{1}{12}$
(c) $\frac{1}{20}$ (d) $\frac{1}{16}$
12. The number of acute angled triangles whose sides are three consecutive positive integers and whose perimeter is at most 100 is
- (a) 28 (b) 29
(c) 31 (d) 33
13. The equation of the straight line passing through the point M(-5,4), such that the portion of it between the axes is divided by the point M is to two equal halves, is
- (a) $10y - 8x = 80$ (b) $8y + 10x = 80$
(c) $10y + 8x = 80$ (d) $8y + 10x + 80 = 0$
14. The value of $\cos^2 \frac{\pi}{8} + \cos^2 \frac{3\pi}{8} + \cos^2 \frac{5\pi}{8} + \cos^2 \frac{7\pi}{8}$ is
- (a) 1 (b) $\frac{3}{2}$
(c) 2 (d) $\frac{9}{4}$
15. If $\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots$ upto $\infty = \frac{\pi^2}{6}$, then value of $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$ upto ∞
- (a) $\frac{\pi^2}{8}$ (b) $\frac{\pi^2}{16}$
(c) $\frac{\pi^2}{36}$ (d) $\frac{\pi^2}{12}$
16. A man is known to speak the truth on an average 4 out of 5 times. He throws a die and reports that it is a five. The probability that it is actually a five is.
- (a) $\frac{4}{9}$ (b) $\frac{5}{9}$
(c) $\frac{4}{15}$ (d) $\frac{2}{15}$
17. If $\log_5 \log_8 (x^2 - 1) = 0$ then a possible value of x is
- (a) $2\sqrt{2}$ (b) $\sqrt{2}$
(c) 2 (d) 3
18. Consider the following statements:
- (a) When $0 < x < 1$, then $\frac{1}{1+x} < 1 - x + x^2$
(b) When $0 < x < 1$, then $\frac{1}{1+x} > 1 - x + x^2$
(c) When $-1 < x < 0$, then $\frac{1}{1+x} < 1 - x + x^2$
(d) When $-1 < x < 0$, then $\frac{1}{1+x} > 1 - x + x^2$
- (a) (i) and (ii) (b) (ii) and (iv)
(c) (i) and (iv) (d) (ii) and (iii)

19. Fifty litres of a mixture of milk and water contains 30 percent of water. This mixture is added to eighty litres of another mixture of milk and water that contains 20 percent of water. Then, how many litres of water should be added to the resulting mixture to obtain a final mixture that contains 25 percent of water?
- (a) 1 (b) 2
(c) 3 (d) 4
20. Three workers working together need 1 hour to construct a wall. The first worker, working alone, can construct the wall twice as fast as the third worker, and can complete the task an hour sooner than the second worker. Then, the average time in hours taken by the three workers, when working along, to construct the wall is
- (a) $(\sqrt{33} + 4)/3$ (b) $(\sqrt{33} + 5)/3$
(c) $(\sqrt{33} + 6)/3$ (d) $(\sqrt{33} + 7)/3$
21. In a class, students are assigned roll numbers from 1 to 140. All students with even roll numbers opted for cricket, all those whose roll numbers are divisible by 5 opted for football, and all those whose roll numbers are divisible by 3 opted for basketball. The number of students who did not opt for any of the three sports is
- (a) 102 (b) 38
(c) 98 (d) 42
22. Given $f(x) = x^2 + \log_3 x$ and $g(y) = 2y + f(y)$, the value of $g(3)$ equals
- (a) 16 (b) 15
(c) 25 (d) 26
23. A 2×2 matrix is filled with four distinct integers randomly chosen from the set $\{1, 2, 3, 4, 5, 6\}$. Then the probability that the matrix generated in such a way is singular is
- (a) $\frac{2}{45}$ (b) $\frac{1}{45}$
(c) $\frac{4}{15}$ (d) $\frac{1}{15}$
24. Ashok started a business with a certain investment. After few months, Bharat joined him investing half amount of Ashok's initial investment. At the end of the first year, the total profit was divided between them in ratio 3:1. Bharat joined Ashok after
- (a) 2 months (b) 3 months
(c) 4 months (d) 6 months
25. The average marks of 6 students in a test is 64. All the students got different marks, one of the student obtained 70 marks and all other students scored 40 or above. The maximum possible difference between the second highest and the second lowest marks is
- (a) 50 (b) 54
(c) 57 (d) 58

Instructions [26 - 30]

The table below presents the quoted buy and sell prices of five stocks during the five trading days of a given week. The quoted sell price is the price at which an investor can sell a stock in the market. The quoted buy price is the price at which an investor can buy a stock from the market. All the quoted numbers are in Indian Rupees

Day	Sunday	Tuesday	Wednesday	Thursday	Friday
Quote	Sell	Buy	Sell	Buy	Sell
Dabur	460	462	455	458	433
Marico	345	346	335	336	365
HUL	1931	1933	1952	1955	1979
ITC	237	238	238	239	246
Britannia	3044	3046	3100	3101	3110

26. If an investor had Rs.36,00,000 to invest in any particular single stock, and she could buy the stock only on Monday and sell it off only on Friday, then the stock she should buy on Monday to earn the maximum possible profit during the week is
- (a) Marico (b) HUL
(c) ITC (d) Britannia
27. If an investor planned to invest Rs. 36,00,000 in purchasing the stocks of HUL on Monday, sell them off on Wednesday and use the entire proceeds to purchase the stocks of Britannia on the same day and sell them off again on Friday, then the total investment return during the week would be
- (a) 2.80 percent (b) 3.00 percent
(c) 3.20 percent (d) 3.40 percent
28. The difference between the quoted buy and sell price of a stock is referred to as the spread of the stock. The average spread of the stocks is lowest on
- (a) Monday (b) Tuesday
(c) Thursday (d) Friday
29. A brokerage firm charges 0.1 percent trading commission on the value of shares bought or sold through its trading platform. If an investor bought 1000 shares of Britannia on Tuesday, and sold all of them on Thursday, then the total brokerage fee that will be charged from the investor is
- (a) 6,125 (b) 6,126
(c) 6,127 (d) 6,128
30. If you had decided to invest Rs. 36,00,000 worth of ITC stocks on Monday, then the day of the week you should choose to sell the stocks to earn the maximum possible profit would be
- (a) Tuesday (b) Wednesday
(c) Thursday (d) Friday

Verbal Ability

Instructions [31-36]

Read the following passage and choose the answer that is closest to each of the questions that are based on the passage.

"John Muir, Earth-planet, Universe." - These words are written on the inside cover of the notebook from which the contents of this volume have been taken. They reflect the mood in which the late author and explorer undertook his thousand-mile walk to the Gulf of Mexico a half-century ago. No less does this refreshingly cosmopolitan address, which might have startled any finder of the book, reveal the temper and the comprehensiveness of Mr. Muir's mind. Even at the early age of twenty-nine his eager interest in every aspect of the natural world had made him a citizen of the universe.

On these expeditions he had disciplined himself to endure hardship, for his notebooks disclose the fact that he often went hungry and slept in the woods, or on the open prairies, with no cover except the clothes he wore.

"Oftentimes," Mr. Muir writes in some unpublished biographical notes, "I had to sleep out without blankets, and also without supper or breakfast. But usually I had no great difficulty in finding a loaf of bread in the widely scattered clearings of the farmers. With one of these big backwoods loaves I was able to wander many a long, wild mile, free as the winds in the glorious forests and bogs, gathering plants and feeding on God's abounding, inexhaustible spiritual beauty bread. Only once in my long Canada wanderings was the deep peace of the wilderness savagely broken. It happened in the maple woods about midnight, when I was cold and my fire was low. I was awakened by the awfully dismal howling of the wolves, and got up in haste to replenish the fire,"

Had it not been for the accidental injury to his right eye in the month of March, 1867, he probably would have started somewhat earlier than he did. In a letter written to Indianapolis friends on the day after the accident, he refers mournfully to the interruption of a long-cherished plan, "For weeks," he writes, "I have daily consulted maps in locating a route through the Southern States, the West Indies, South America, and Europe - a botanical journey studied for years. But, alas, I am half blind. My right eye, trained to minute analysis, is lost and I have scarce heart to open the other."

The injury to his eye proved to be less serious than he had at first supposed. In June he was writing to a friend: "I have been reading and botanizing for some weeks, and find that for such work I am not very much disabled."

In an account written after the excursion he says: "I was eager to see Illinois prairies on my way home, so we went to Decatur, near the center of the State, thence, I botanized one week on the prairie about seven miles southwest of Pecatonica.... To me all plants are more precious than before. My poor eye is not better, nor worse, A cloud is over it, but in gazing over the widest landscapes, I am not always sensible of its presence."

31. "John Muir, Earth-planet, Universe," Muir wrote this in his notebook because
- (a) He did not have a permanent address.
 - (b) He wanted to be a traveler all his life.
 - (c) His love for botany went beyond borders.
 - (d) He wanted to live in the open all his life.
32. Which of these did John Muir have no great difficulty in doing?
- (a) Finding food among the farmers.
 - (b) Bating regularly every day.
 - (c) Sleeping without blankets on certain nights.
 - (d) Chasing wolves that howled through the nights.
33. How did the experience with nature affect John Muir?
- (a) He felt spiritually fulfilled.
 - (b) He became a better botanist.
 - (c) He felt successful after collecting rare plants.
 - (d) He enjoyed the breeze in the glorious forests.

34. According to the passage, in what way was Muir affected by his injury?
- (a) His injured eye healed well very slowly.
 - (b) He could not open his left eye during his exploration.
 - (c) His eyesight was affected, but he was able to carry on.
 - (d) He took a long time to consult maps and study them.
35. When Muir said that lie 'botanized', he meant that he
- (a) studied plants and made notes on them.
 - (b) studied plants in their natural habitat
 - (c) studied plants during his exploration.
 - (d) studied plants and classified them.
36. Which of these sentences is not true?
- (a) John Muir explored several new places even on his way home.
 - (b) There was a cloud over the landscape he travelled
 - (c) He was commuted to being a botanist more than ever.
 - (d) He recorded his adventures in the notebook.

Instructions [37-42]

Complete the following sentences by choosing the appropriate word/phrase from the options given below.

37. It came as a shock to me that my friend had lost the ____ he had built carefully over the years when in a moment of madness he lent all of it to his neighbour.
- (a) nest egg
 - (b) nests and eggs
 - (c) bird's nest
 - (d) nestled eggs
38. During the debate, when the minister appeared to be vague about his stand on the controversial energy his opponent asked him to _____ and declare outright if the government would go ahead with the project or not.
- (a) be curt to the face
 - (b) cut to the chase
 - (c) curb the farce
 - (d) cut globe
39. It is irritating when co-workers dismiss your ideas as worthless, and later, present those very ideas as their own. They just want to _____
- (a) steal your credit
 - (b) steal your crown
 - (c) steal your thunder
 - (d) steal your gloating
40. Before the start of the meeting, I am _____ on the latest developments,
- (a) taking you through
 - (b) going to share
 - (c) going to bring you up to speed
 - (d) going to be explaining
41. Despite statistics to prove that there was a slowdown in the economy of the country, the professor was _____ about it.
- (a) in affirmation
 - (b) in denial
 - (c) in protest
 - (d) in shock

42. After the unearthing of relics at the archaeological site, the historian was asked to _____ the significance of the discovery.
- (a) throw light on (b) shed meaning on
(c) turn light on (d) focus meaning on

Instructions [43-45]

Choose the alternative so that the underlined part of the sentence is rendered correct.

43. The pharmaceutical company hired a consultant to supervise a taskforce studying lower salaries as to their effects on employees' morale
- (a) studying what the effects lower salaries would have on employees' morale
 - (b) studying the effects of lower salaries on employee morale
 - (c) for studying what are the effects in employees' morale that lower salaries would cause.
 - (d) studying the effects of employee morale on lower salaries
44. Besides offering physiological rewards such as toned muscles, when regular karate is practiced the body can be turned into a dangerous weapon.
- (a) one can turn the body into a dangerous weapon through karate, if it is practiced regularly
 - (b) the body can be turned into a dangerous weapon as a result of karate if practiced regularly
 - (c) when karate is practiced regularly, the body can be turned into a dangerous weapon
 - (d) karate, if practiced regularly, can turn the body into a dangerous weapon
45. The animal activist has raised awareness not only on the plight of abandoned dogs, but also on overworked bulls and temple elephants.
- (a) on the plight of not only the abandoned dogs, but also on those of overworked bulls
 - (b) not only on the plight of the abandoned dogs, but also on that of overworked bulls
 - (c) the plight of abandoned dogs, as well as the over worked bulls
 - (d) the plight of abandoned dogs, but also on overworked bulls

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Instructions [46-47]

Each of the paragraphs given below has a sentence missing which is indicated by a blank. From the choices given below each paragraph, choose the sentence that seems most logically appropriate to complete the paragraph.

46. When people move from one city or country to another, the spread of diseases may result. People often bring in germs which may not have been present there before. These new germs can spread quickly and cause previously unknown diseases. _____ They become ill more easily and could even die. In turn, newcomers may catch diseases which were not present where they came from.
- (a) If they had gone back, they would have started an epidemic.
 - (b) Pollution also can lead to the spread of disease.
 - (c) If a germ is completely new to a region, people have no natural protection against it.
 - (d) Such changes may result in enhancing conditions for people who live in big cities.
47. Samar, a young man from a village in Tripura is a social media star who hopes to make it big as a model and actor. While his minute-long videos are very popular, his series titled, "Who wore it better?" showcases his creative side. He uses a variety of filters and visual effects to improve his videos. _____

47. Samar, a young man from a village in Tripura is a social media star who hopes to make it big as a model and actor. While his minute-long videos are very popular, his series titled, "Who wore it better?" showcases his creative side. He uses a variety of filters and visual effects to improve his videos. _____

- (a) It is clear that short videos are very popular among the youth than television.
- (b) This is one of the ways young men make money on the internet.
- (c) it is possible to be popular on the internet, although making money here is difficult.
- (d) This shows how it is possible today for people with limited resources to express their creativity.

Instructions [48-52]

The sentences below have words that are missing.

Choose the best option from those given below to complete the sentence.

48. Coral reefs are one of the most _____ biologically complex, and diverse marine ecosystems on Earth. This ecosystem is one of the _____ paradoxes of the biosphere: how do clear, and thus nutrient-poor, waters support such _____ and productive communities?
- (a) common, sensitive, fruitful
 - (b) fragile, fascinating, prolific
 - (c) deep, strange, exuberant
 - (d) frail, unrecognised, wealthy live in a land without
49. It is a _____ live in a land without _____. Even with all the technological advancement and economic growth, there are still millions who live in war zones where life every day is filled with _____.
- (a) coincidence, prejudice, adventures
 - (b) privilege, bigotry, terror
 - (c) lovely, pride, episodes
 - (d) punishment, hate, agony
50. The ability to _____ and _____ a cohesive team is particularly critical in hi-tech firms where the _____ landscape can shift dramatically in the face of disruptive technologies.
- (a) mentor, keep, sustainable
 - (b) shape, guide, competitive
 - (c) drive, maintain, tectonic
 - (d) involve, manage, green
51. Cool-headed and rational, Jo Eun-cha is a talented and driven anchorman who commands as well as demands respect. After _____ to get elected as a news director, he aims to _____ to his job as the top news anchor to increase his _____ of being elected the next year.
- (a) happening, quit, hopes
 - (b) failing, return, chances
 - (c) hoping, upgrade, certainty
 - (d) going, give up, luck
52. A collective investment fund (CIF), also known as a collective investment trust (CIT), is a group of _____ accounts held by a bank or trust company. The financial institution assets from individuals and organizations to _____ a single larger, diversified portfolio.
- (a) diverse, discounts, conceive
 - (b) different, collected, simulate
 - (c) financial, covered, synthesise
 - (d) pooled, groups, develop
53. One of the statements below contains a word used incorrectly. Choose the option which has the incorrect or inappropriate usage of the word.
- (a) The President of the City Chamber of Commerce is an urbane, kindly and generous man.
 - (b) The writer's next book deals with the life of an urbane intellectual as seen through the eyes of a lawyer,
 - (c) My uncle who lives in New York turned out to be the exact opposite of the urbane, wealthy gentleman that I was expecting to meet,
 - (d) Over 70% of the people living in urbane areas said that they wanted better roads and more schools.

54. One of the statements below contains a word used incorrectly. Choose the option which has the incorrect or inappropriate usage of the word.
- (a) The young scientist built a small, low-cost machine to recycle waste water. It was a simple but ingenious solution to die problem of water shortage in the city.
 - (b) My friend is an ingenious cook-she can make very delicious dishes from the most ordinary ingredients.
 - (c) The young man lost all his savings because he was rather ingenious to believe the agent's promises of a well-paid job and a comfortable life in a foreign country,
 - (d) Teachers say that some children are very ingenious when it comes to finding excuses for not completing their work.
55. One of the statements below contains a word used incorrectly. Choose the option which has the incorrect or inappropriate usage of the word.
- (a) After several rounds of meetings, the city council passed an ordinance to limit the number of cafes near the seaside.
 - (b) As per a royal ordinance, all men under 18 had to join die army and fight the war.
 - (c) Military aircrafts can he used to curry a wide variety of ordinance.
 - (d) The people pro tested a city ordinance that all parks must be closed by 9 pm.
56. One of the statements below contains a word used incorrectly. Choose the option which has the incorrect or inappropriate usage of the word.
- (a) It is a pleasure when we are all together for a meal at home.
 - (b) Her son is not interested in watching movies, which is not all together a bad thing.
 - (c) The three hundred guests sat all together in the large dining hall of the hotel.
 - (d) The facilitator asked the participant to stand up, all together.
57. One of the statements below contains a word used incorrectly. Choose the option which has the incorrect or inappropriate usage of the word.
- (a) The sickness of a family member can affect the lives of everyone at home.
 - (b) Kerala was affected by severe Hood all through the year.
 - (c) Taking this medicine for a prolonged period can have severe after affects.
 - (d) The popular star was not a natural actor, his style was rather affected.

Instructions [58-60]

The sentences given below, when properly sequenced, form a coherent paragraph. Each sentence is labelled with a number. Decide on the most logical order and enter the sequence of numbers in the space provided.

58. (1) However, they stand out because they help the film maker realize his personal artistic vision.
(2) An Indie film refers to independent films made outside a major studio system.
(3) Their release is also limited to a few screens, although there have been films with a wide release.
(4) Most of such dims arc low budget films produced and distributed by small companies.
- (a) 2431 (b) 2412
(c) 2314 (d) 3124

59. (1) Mental illness continues to struggle for recognition as a significant contributor to poor health in spite of the existence of the Mental Healthcare Act 2017.
- (2) The data reveals that an estimated one in seven Indians-or 197 million persons- suffered from mental disorders, at varying degrees of severity, in 2017.
- (3) There is always more to human health than meets the eye.
- (4) This is particularly important in the Indian cultural and political context, where the outward appearance of physical well-being is often considered the only visible marker of overall health.
- (a) 3214 (b) 3412
- (c) 2431 (d) 4312
60. (1) It was felt that the major polluters had a moral obligation to deliver on the agreements reached some years ago during a similar convention on Climate Change.
- (2) The papers presented indicated that the impacts of current warming are much more severe than previously understood.
- (3) It appears that the planet's ability to adapt to and cope with how it is being treated is fraying.
- (4) For example, the acceleration of sea level rise and ocean warming, and increasing risks of reaching limits to adaptation.
- (5) This is what emerged at a recent conference attended by the largest scientific community on climate science.
- (a) 25413 (b) 15243
- (c) 34521 (d) 35241

A Place to Learn, A Place to Grow

Answer Key Quantitative Ability SA

1. **Answer:** 34

Explanation: We use the division algorithm:

$$\text{Dividend} = \text{Divisor} \times \text{Quotient} + \text{Remainder}$$

Given:

- Divisor = 5
- $Q \times R = 24$
- $Q + R = 10$

Using hit and trial:

$$Q = 6, R = 4$$

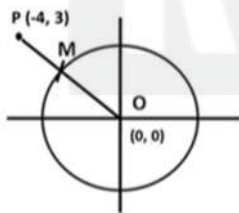
(Remainder must be less than the divisor.)

Now,

$$n = 5 \times 6 + 4 = \boxed{34}$$

2. **Answer:** 4 units

Explanation:



Given equation of circle:

$$x^2 + y^2 = 1^2$$

Centre = (0, 0), Radius = 1

Point $P(-4, 3)$

Distance PO :

$$PO = \sqrt{(-4)^2 + 3^2} = \sqrt{16 + 9} = 5$$

Shortest distance from point to circle:

$$PM = PO - r = 5 - 1 = \boxed{4}$$

3. **Answer:** 16

Explanation: Given:

$$0.04 \log_{\sqrt{5}} \left(\frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \right)$$

Sum of GP:

$$\frac{1}{4} \left(\frac{1}{1 - \frac{1}{2}} \right) = \frac{1}{2}$$

So expression becomes:

$$0.04 \log_{\sqrt{5}} \left(\frac{1}{2} \right)$$

$$= \frac{4}{100} \times \frac{\log(1/2)}{\log(\sqrt{5})} = \frac{1}{25} \times \frac{2 \log(1/2)}{\log 5}$$

$$= \frac{2}{25} \log_5 \left(\frac{1}{2} \right) = \log_5 (16)$$

$$= \boxed{16}$$

4. **Answer:** $abc = 1$

Explanation: Given determinant:

$$\begin{vmatrix} a & a^2 & a^3 - 1 \\ b & b^2 & b^3 - 1 \\ c & c^2 & c^3 - 1 \end{vmatrix} = 0$$

Split using column addition:

$$\begin{vmatrix} a & a^2 & a^3 \\ b & b^2 & b^3 \\ c & c^2 & c^3 \end{vmatrix} - \begin{vmatrix} a & a^2 & 1 \\ b & b^2 & 1 \\ c & c^2 & 1 \end{vmatrix}$$

Factorising:

$$= abc \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} - \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}$$

$$(abc - 1) \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = 0$$

Since a, b, c are distinct, determinant $\neq 0$.

$$\Rightarrow abc = \boxed{1}$$

5. **Answer:** 7

Explanation: Given:

$$f(x) = |3 - x| + |2 + x| + |5 - x|$$

Critical points:

$$x = -2, 3, 5$$

Arrange:

$$-2 < 3 < 5$$

Minimum value occurs at the middle critical point:

$$x = 3$$

$$f(3) = |0| + |5| + |2| = \boxed{7}$$

6. **Answer:** ₹100

Explanation: Let:

- Cost of pen = A
- Cost of pencil = B

Second day:

$$4A + B = 112$$

First day (ratio 2:3):

$$2A + 3B = 86$$

Solving:

$$A = 25$$

Cost of 4 pens:

$$4 \times 25 = \boxed{\text{₹100}}$$

7. **Answer:** 7920

Explanation: Let the number be $ABCD$.

Given:

- $A \times D = 0 \Rightarrow D = 0$
- $A - D = 7 \Rightarrow A = 7$

- $B \times C = 18$
- $B - C = 7$

Only possible pair:

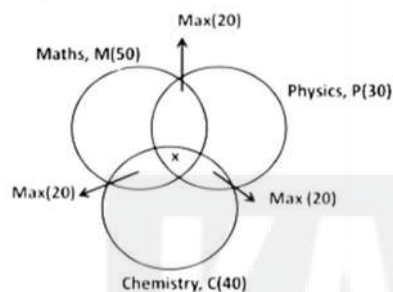
$$B = 9, C = 2$$

Number formed:

7920

8. **Answer:** 20

Explanation:



Using inclusion-exclusion:

$$n(M \cap P \cap C) = n(M \cup P \cup C) + [n(MP) + n(MC) + n(PC)] - [n(M) + n(P) + n(C)]$$

Substituting maximum values:

$$80 + (20 + 20 + 20) - (50 + 30 + 40) = \boxed{20}$$

9. **Answer:** 3

Explanation: Total race length = 3000 m

Speed ratio = 3 : 2

Let speeds = $3x$ and $2x$

Meeting interval:

$$\text{LCM}\left(\frac{300}{3x}, \frac{300}{2x}\right) = \frac{300}{x}$$

Distance covered by faster runner per meeting:

$$3x \times \frac{300}{x} = 900$$

Before completing 3000 m:

900, 1800, 2700

Thus, meetings = 3

10. **Answer:** 256

Explanation:

	Distinct Objects (9)	Number of ways of selection
Case I	Choose 4	9C_4 ways
Case II	Choose 3	9C_3 ways
Case III	Choose 2	9C_2 ways
Case IV	Choose 1	9C_1 ways
Case V	Choose 0	9C_0 ways

×

×

×

×

×

Identical Objects (4)	Number of ways of selection
Choose 0	1 way
Choose 1	1 way
Choose 2	1 way
Choose 3	1 way
Choose 4	1 way

Selecting 4 objects where:

- 9 are distinct
- 4 are identical

Total ways:

$$\sum_{k=0}^4 {}^9C_k$$
$$= 1 + 9 + 36 + 84 + 126 = \boxed{256}$$



A Place to Learn, A Place to Grow

Quantitative Ability MCQ

11. **Answer:** D

Explanation: $10^{19} = 5^{19} \times 2^{19}$

Total factors:

$$(19 + 1)(19 + 1) = 400$$

Factors divisible by 10^5 :

$$(4 + 1)(4 + 1) = 25$$

Probability:

$$\frac{25}{400} = \boxed{\frac{1}{16}}$$

12. **Answer:** B

Explanation: Acute triangle condition:

$$c^2 < a^2 + b^2$$

Possible consecutive triplets:

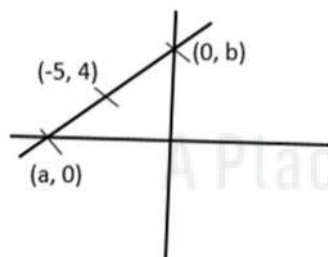
$$(1,2,3) \text{ to } (32,33,34) \Rightarrow 32 \text{ sets}$$

Eliminating first 3 (not acute):

$$32 - 3 = \boxed{29}$$

13. **Answer:** A

Explanation:



Midpoint given as $(-5, 4)$

Using midpoint formula:

$$\frac{a+0}{2} = -5 \Rightarrow a = -10$$

$$\frac{b+0}{2} = 4 \Rightarrow b = 8$$

Equation of line:

$$\frac{y}{x+10} = \frac{8}{10} \Rightarrow 10y - 8x = 80$$

Correct option is (a).

14. **Answer:** C

Explanation: $\cos^2 \frac{\pi}{8} + \cos^2 \frac{3\pi}{8} + \cos^2 \frac{5\pi}{8} + \cos^2 \frac{7\pi}{8}$

Using identity:

$$\cos^2 \theta = \cos^2(\pi - \theta)$$

$$= 2 \left(\cos^2 \frac{\pi}{8} + \cos^2 \frac{3\pi}{8} \right)$$

Since angles are complementary:

$$\cos^2 A + \cos^2 B = 1$$

$$\Rightarrow \boxed{2}$$

15. **Answer: A**

Explanation: Given:

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots = \frac{x^2}{6}$$

Even term series:

$$\frac{1}{2^2} \left(1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots \right) = \frac{x^2}{24}$$

Odd term series:

$$\frac{x^2}{6} - \frac{x^2}{24} = \boxed{\frac{x^2}{8}}$$

16. **Answer: A**

Explanation: Let

- S_1 : Man speaks the truth
- S_2 : Man speaks a lie
- E : Five appears on the die

Given:

$$P(S_1) = \frac{4}{5}, P(S_2) = \frac{1}{5}$$

If the man speaks truth:

$$P(E | S_1) = \frac{1}{6}$$

If the man lies:

$$P(E | S_2) = 1 - \frac{1}{6} = \frac{5}{6}$$

Using Bayes' Theorem:

$$\begin{aligned} P(S_1 | E) &= \frac{P(S_1)P(E|S_1)}{P(S_1)P(E|S_1) + P(S_2)P(E|S_2)} \\ &= \frac{\frac{4}{5} \times \frac{1}{6}}{\frac{4}{5} \times \frac{1}{6} + \frac{1}{5} \times \frac{5}{6}} = \frac{\frac{4}{30}}{\frac{4}{30} + \frac{1}{30}} = \frac{4}{5} \end{aligned}$$

17. **Answer: D**

Explanation: Given:

$$\log_5(\log_8(x^2 - 1)) = 0$$

$$\text{Since } \log a = 0 \Rightarrow a = 1,$$

$$\log_8(x^2 - 1) = 1$$

$$x^2 - 1 = 8^1 = 8 \Rightarrow x^2 = 9 \Rightarrow x = \pm 3$$

From given options, valid **Answer:**

$$\boxed{x = 3}$$

18. **Answer: C**

Explanation: We test values as inequalities are involved.

Test for statements (i) & (ii):

$$\text{Let } x = \frac{1}{2}$$

$$\frac{1}{1+x} = \frac{1}{1+\frac{1}{2}} = \frac{2}{3}$$

$$1 - x + x^2 = 1 - \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$$

$$\frac{2}{3} < \frac{3}{4} \Rightarrow (i) \text{ true, } (ii) \text{ false}$$

Test for statements (iii) & (iv):

Let $x = -\frac{1}{2}$

$$\frac{1}{1+x} = 2, 1 - x + x^2 = \frac{7}{4}$$

$$2 > \frac{7}{4} \Rightarrow (iv) \text{ true, } (iii) \text{ false}$$

Correct statements: (i) and (iv)

(c)

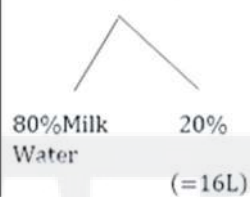
19. **Answer: B**

Explanation:

Mixture 1 (50L)



Mixture 2 (80L)



Total solution = 130L

Water already present:

$$15 + 16 = 31 \text{ L}$$

Let x L water be added.

Final condition:

Water = 25% of total solution

$$31 + x = \frac{1}{4}(130 + x)$$

$$124 + 4x = 130 + x \Rightarrow 3x = 6 \Rightarrow x = \boxed{2 \text{ L}}$$

20. **Answer: A**

Explanation: Let first worker take t hours.

Then:

- Second worker = $t + 1$
- Third worker = $2t$

Combined work:

$$\frac{1}{t} + \frac{1}{t+1} + \frac{1}{2t} = 1$$

Solving:

$$t = \frac{3 + \sqrt{33}}{4}$$

Average time:

$$\frac{t + (t+1) + 2t}{3} = \boxed{\frac{\sqrt{33} + 4}{3}}$$

21. **Answer: B**

Explanation: Total students = 140

Cricket (even rolls):

$$\left\lfloor \frac{140}{2} \right\rfloor = 70$$

Football (multiple of 5):

$$\left\lfloor \frac{140}{5} \right\rfloor = 28$$

Basketball (multiple of 3):

$$\left\lfloor \frac{140}{3} \right\rfloor = 46$$

Intersections:

- $C \cap F(10)$: 14
- $C \cap B(6)$: 23
- $F \cap B(15)$: 9
- All three (30): 4

Using inclusion-exclusion:

$$n(C \cup F \cup B) = 70 + 28 + 46 - 14 - 23 - 9 + 4 = 102$$

Students not playing any sport:

$$140 - 102 = \boxed{38}$$

22. **Answer:** A

Explanation: $f(x) = x^2 + \log_3 x$, $g(y) = 2y + f(y)$

$$g(3) = 2(3) + f(3) = 6 + f(3)$$

$$f(3) = 9 + \log_3 3 = 9 + 1 = 10$$

$$g(3) = 6 + 10 = \boxed{16}$$

23. **Answer:** B

Explanation: Matrix:

$$A = \begin{vmatrix} a & c \\ d & b \end{vmatrix}$$

For singular matrix:

$$|A| = ab - cd = 0 \Rightarrow ad = bc$$

Total outcomes:

$$6C4 \times 4! = 6P4 = \frac{6!}{2!} = 360$$

Valid sets satisfying $ad = bc$:

- $\{1, 2, 3, 6\}$
- $\{2, 3, 4, 6\}$

Each gives 4 arrangements.

Favourable outcomes = 8

$$P = \frac{8}{360} = \boxed{\frac{1}{45}}$$

24. **Answer:** C

Explanation: Profit $\propto$ Investment $\times$ Time

$$\frac{P_A}{P_B} = \frac{I_A \times 12}{\frac{I_A}{2} \times x} = \frac{12}{x/2}$$

$$\frac{3}{1} = \frac{24}{x} \Rightarrow x = 8 \text{ months}$$

Bharat joined after:

$$12 - 8 = \boxed{4 \text{ months}}$$

25. **Answer:** B

Explanation: Average of 6 students = 64

$$\text{Total} = 6 \times 64 = 384$$

Let five distinct scores be:

$$A < B < C < D < E$$

Given:

$$A + B + C + D + E + 70 = 384 \Rightarrow A + B + C + D + E = 314$$

To maximize $D - B$, minimize A, B, C :

$$A = 40, B = 41, C = 42$$

$$D + E = 314 - 123 = 191$$

Closest split:

$$D = 95, E = 96$$

$$D - B = 95 - 41 = \boxed{54}$$

Instructions [26 - 30]

The table below presents the quoted buy and sell prices of five stocks during the five trading days of a given week. The quoted sell price is the price at which an investor can sell a stock in the market. The quoted buy price is the price at which an investor can buy a stock from the market. All the quoted numbers are in Indian Rupees

Day	Sunday	Tuesday	Wednesday	Thursday	Friday
Quote	Sell	Buy	Sell	Buy	Sell
Dabur	460	462	455	458	433
Marico	345	346	335	336	365
HUL	1931	1933	1952	1955	1979
ITC	237	238	238	239	246
Britannia	3044	3046	3100	3101	3110

26. **Answer: A**

Explanation: Investment = ₹36 lakh

Compute return ratios:

- Marico: $372/346 = 1.07$
- HUL: $1966/1933 = 1.01$
- ITC: $253/238 = 1.06$
- Britannia: $3140/3046 = 1.03$

Maximum return:

Marico

27. **Answer: C**

Explanation: HUL shares bought Monday:

$$\frac{36 \text{ lakh}}{1933}$$

Sold Wednesday at ₹1979.

Britannia shares bought Wednesday at ₹3115.

Sold Friday at ₹3140.

Final amount:

₹37.15 lakh

$$\% \text{ return} = \frac{37.15 - 36}{36} \times 100 = \boxed{3.19\%}$$

28. **Answer:** A

Explanation: Since individual spreads of stocks are lowest on Monday, their average spread will also be minimum on Monday.

Monday

29. **Answer:** B

Explanation: Investment (Tuesday):

$$1000 \times 3101 = ₹31,01,000$$

Selling (Thursday):

$$1000 \times 3025 = ₹30,25,000$$

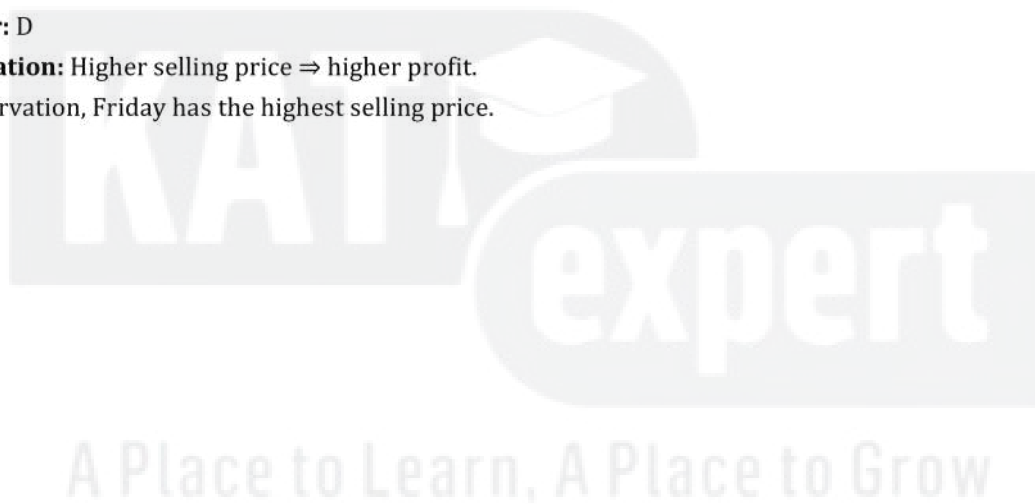
Brokerage = 0.1% of total turnover:

$$0.001(31,01,000 + 30,25,000) = ₹6126$$

30. **Answer:** D

Explanation: Higher selling price $\Rightarrow$ higher profit.
By observation, Friday has the highest selling price.

Friday



Verbal Ability

Instructions [31-36]

Read the following passage and choose the answer that is closest for each of the questions that are based on the passage.

"John Muir, Earth-planet, Universe." - These words are written on the inside cover of the notebook from which the contents of this volume have been taken. They reflect the mood in which the late author and explorer undertook his thousand-mile walk to the Gulf of Mexico a half-century ago. No less does this refreshingly cosmopolitan address, which might have startled any finder of the book, reveal the temper and the comprehensiveness of Mr. Muir's mind. Even at the early age of twenty-nine his eager interest in every aspect of the natural world had made him a citizen of the universe.

On these expeditions he had disciplined himself to endure hardship, for his notebooks disclose the fact that he often went hungry and slept in the woods, or on the open prairies, with no cover except the clothes he wore.

"Often times," Mr. Muir writes in some unpublished biographical notes, "I had to sleep miles without blankets, and also without supper or breakfast. But usually I had no great difficulty in finding a loaf of bread in the widely scattered clearings of the farmers. With one of these big backwoods loaves I was able to wander many a long, wild mile, free as the winds in the glorious forests and bogs, gathering plants and feeding on God's abounding, inexhaustible spiritual beauty bread. Only once in my long Canada wanderings was the deep peace of the wilderness savagely broken. It happened in the maple woods about midnight, when I was cold and my fire was low. I was awakened by the awfully dismal howling of the wolves, and got up in haste to replenish the fire,"

Had it not been for the accidental injury to his right eye in the month of March, 1867, he probably would have started somewhat earlier than he did. In a letter written to Indianapolis friends on the day after the accident, he refers mournfully to the interruption of a long-cherished plan, "For weeks," he writes, "I have daily consulted maps in locating a route through the Southern States, the West Indies, South America, and Europe - a botanical journey studied for years. But, alas, I am half blind. My right eye, trained to minute analysis, is lost and I have scarce heart to open the other."

The injury to his eye proved to be less serious than he had at first supposed. In June he was writing to a friend: "I have been reading and botanizing for some weeks, and find that for such work I am not very much disabled."

In an account written after the excursion he says: "I was eager to see Illinois prairies on my way home, so we went to Decatur, near the center of the State, thence, I botanized one week on the prairie about seven miles southwest of Pecatonica.... To me all plants are more precious than before. My poor eye is not better, nor worse, A cloud is over it, but in gazing over the widest landscapes, I am not always sensible of its presence."

31. **Answer: C**

Explanation: Options (a), (b), and (d) can be eliminated as the passage does not mention any of these ideas either directly or indirectly. Refer to the last line of the first paragraph ("...citizen of the universe"), which clearly indicates that Muir's love for nature and botany extended beyond geographical boundaries.

32. **Answer: A**

Explanation: This is a straightforward question with a direct answer. Refer to the second line of the third paragraph, which clearly states that Muir had no great difficulty in finding food from farmers.

33. **Answer: A**

Explanation: Option (b) can be ruled out as the passage does not state that Muir became a better botanist. Option (c) is incorrect since there is no mention of success in collecting rare plants. Option (d) distorts the idea. Option (a) best fits the passage, as indicated in the fourth and fifth lines of the third paragraph, which reflect his spiritual fulfillment.

34. **Answer:** D

Explanation: The final part of the fourth paragraph shows that the injury to his right eye prevented him from consulting maps the way he used to earlier, thereby affecting his plans.

35. **Answer:** B

Explanation: Options (a) and (d) can be eliminated because the passage does not mention Muir making notes or classifying plants. Option (c) is weaker, as it suggests exploration first and study later, whereas the passage implies that his love for nature led him to exploration. Hence, option (b) is the most appropriate.

36. **Answer:** B

Explanation: Option (d) is true, as mentioned clearly in the second paragraph where Muir records his experiences in his notebook. Option (c) is not supported by the passage. The phrase “a cloud is over it” is metaphorical, meaning worry or concern, not an actual cloud. Therefore, option (b) is the incorrect statement.

Instructions [37-42]

Complete the following sentences by choosing the appropriate word/phrase from the options given below.

37. **Answer:** A

Explanation: A *nest egg* refers to money that has been carefully saved over time, usually for future use.

38. **Answer:** B

Explanation: The phrase *cut to the chase* means to come directly to the main point without unnecessary details.

39. **Answer:** C

Explanation: *Steal your thunder* means taking credit for someone else's ideas or achievements.

40. **Answer:** C

Explanation: The phrase *bring you up to speed* means to provide the latest or most relevant information.

41. **Answer:** B

Explanation: The word *despite* signals a contrast. Since the data shows an economic slowdown, the professor refusing to accept it would be *in denial*, making option (b) the correct choice.

42. **Answer:** A

Explanation: *Throw light on* means to explain or clarify something. Options (b) and (d) are grammatically incorrect, while option (c) refers to physically switching on a light.

Instructions [43-45]

Choose the alternative so that the underlined part of the sentence is rendered correct.

43. **Answer:** B

Explanation: Option (a) violates parallelism. Option (d) reverses the meaning, and option (c) is grammatically incorrect. Hence, option (b) is the most accurate and grammatically correct choice.

44. **Answer:** D

Explanation: The word *besides* means “in addition to” and requires a proper subject. Options (a), (b), and (c) fail to maintain grammatical balance. Option (d) corrects the structure appropriately.

45. **Answer: B**

Explanation: The construction *not only... but also* requires parallel structure. Since awareness is raised *of the plight*, it must be followed by *that of* in the second clause. Therefore, option (b) is correct.

Instructions [46-47]

Each of the paragraphs given below has a sentence missing which is indicated by a blank. From the choices given below each paragraph, choose the sentence that seems most logically appropriate to complete the paragraph.

46. **Answer: C**

Explanation: Option (c) logically connects both sides of the paragraph by explaining how new germs affect people who lack immunity, smoothly linking the preceding and following sentences.

47. **Answer: D**

Instructions [48-52]

The sentences below have words that are missing.

Choose the best option from those given below to complete the sentence.

48. **Answer: B**

Explanation: Options (a) and (d) can be eliminated as *fruitful* and *wealthy* do not fit the context. Option (c) is incorrect because “deep” does not suit the first blank. Hence, option (b) fits best.
(*Fragile* = easily damaged; *fascinating* = extremely interesting; *prolific* = abundant.)

49. **Answer: B**

Explanation: Option (a) does not fit contextually, as *adventures* is inappropriate. Options (c) and (d) are also unsuitable. Option (b) fits all blanks logically.
(*Privilege* = advantage; *bigotry* = unreasonable beliefs; *terror* = extreme fear.)

50. **Answer: B**

Explanation: Options (c) and (d) can be eliminated since *tectonic* and *green* do not suit the context. Option (a) is also incorrect as *sustainable landscape* does not shift dramatically. Option (b) is the most suitable.
(*Shape* = form; *guide* = advise; *competitive* = involving competition.)

51. **Answer: B**

Explanation: Option (a) is grammatically incorrect. Option (d) does not fit logically. Between (b) and (c), *chances* fits better than *certainty* in the final blank. Hence, option (b).

52. **Answer: D**

Explanation: The passage focuses on creativity and expression, not monetisation or television. Therefore, options (a), (b), and (c) can be eliminated. Option (d) best completes the paragraph.

53. **Answer: D**

Explanation: The word *urbane* refers to refined behaviour, not areas or regions. Options (a), (b), and (c) use it correctly for people. Option (d) should have used *urban* instead.

54. **Answer:** C

Explanation: *Ingenious* means clever or inventive, while *ingenuous* means innocent or naive. Option (c) incorrectly uses *ingenious*.

55. **Answer:** C

56. **Answer:** B

57. **Answer:** C

Instructions [58-60]

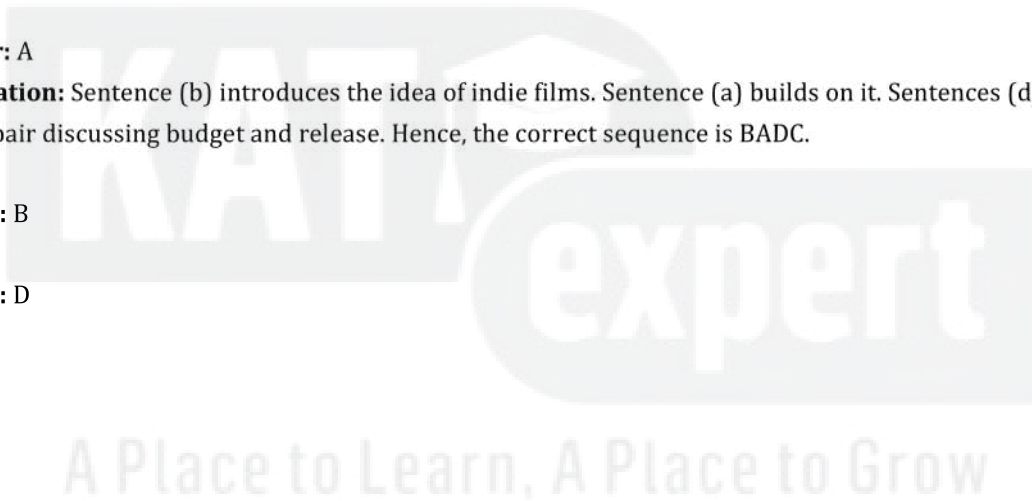
The sentences given below, when properly sequenced, form a coherent paragraph. Each sentence is labelled with a number. Decide on the most logical order and enter the sequence of numbers in the space provided.

58. **Answer:** A

Explanation: Sentence (b) introduces the idea of indie films. Sentence (a) builds on it. Sentences (d) and (c) form a logical pair discussing budget and release. Hence, the correct sequence is BADC.

59. **Answer:** B

60. **Answer:** D



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